

LAB PROJECT REPORT

Objective :

The primary objective of this project is to design a reconfigurable graphene-based microstrip patch antenna for terahertz (THz) wireless communication. THz frequencies, typically ranging from 0.1 to 10 THz, are known for their ability to handle high data transmission rates. The challenge in developing effective antennas at THz frequencies lies in maintaining high radiation efficiency, gain, and tunability. This project aims to overcome these limitations by using graphene, whose surface conductivity can be controlled through an external bias electric field. By dynamically adjusting the chemical potential of graphene, the proposed antenna can tune its operating frequency between 695.55 GHz and 705.05 GHz, achieving high radiation efficiency and gain without degrading performance.

Theoretical Background :

Terahertz communication has become increasingly important due to its high capacity and directional communication capabilities. However, achieving efficient radiation in this band is challenging due to limitations in traditional materials, which suffer from high ohmic losses and limited tunability. Graphene has emerged as a promising material for THz applications due to its exceptional electrical, thermal, and mechanical properties. It can support high-frequency waves and its surface conductivity can be adjusted by varying the chemical potential, offering dynamic tunability.

The surface conductivity of graphene, essential for frequency tuning, is calculated using the Kubo formula, which accounts for in-band and inter-band transitions. In the THz band, the in-band transition typically dominates, and the Drude model can approximate its contribution to surface conductivity:

$$\sigma_{\text{intra}}(\omega, T, \mu_c) = \frac{-ie^2 k_B T}{\pi \hbar^2 (\omega - i\Gamma)} \left(\frac{\mu_c}{k_B T} + 2 \ln \left(e^{-\mu_c/k_B T} + 1 \right) \right)$$

where:

- σ_{intra} is the in-band conductivity,
- ω is the angular frequency,
- T is the temperature,
- μ_c is the chemical potential,
- e is the electron charge, and
- $k_B, \hbar$ are the Boltzmann and reduced Planck constants, respectively.

The proposed antenna can achieve tunable frequencies by dynamically tuning μc via an applied bias. The equivalent circuit model of the antenna includes the graphene patch as a variable admittance, connected in parallel with the transmission line and coupling capacitor. The Drude model predicts the admittance behavior as a function of μc , essential for adjusting the resonant frequency.

Block Diagram Components and Functions:

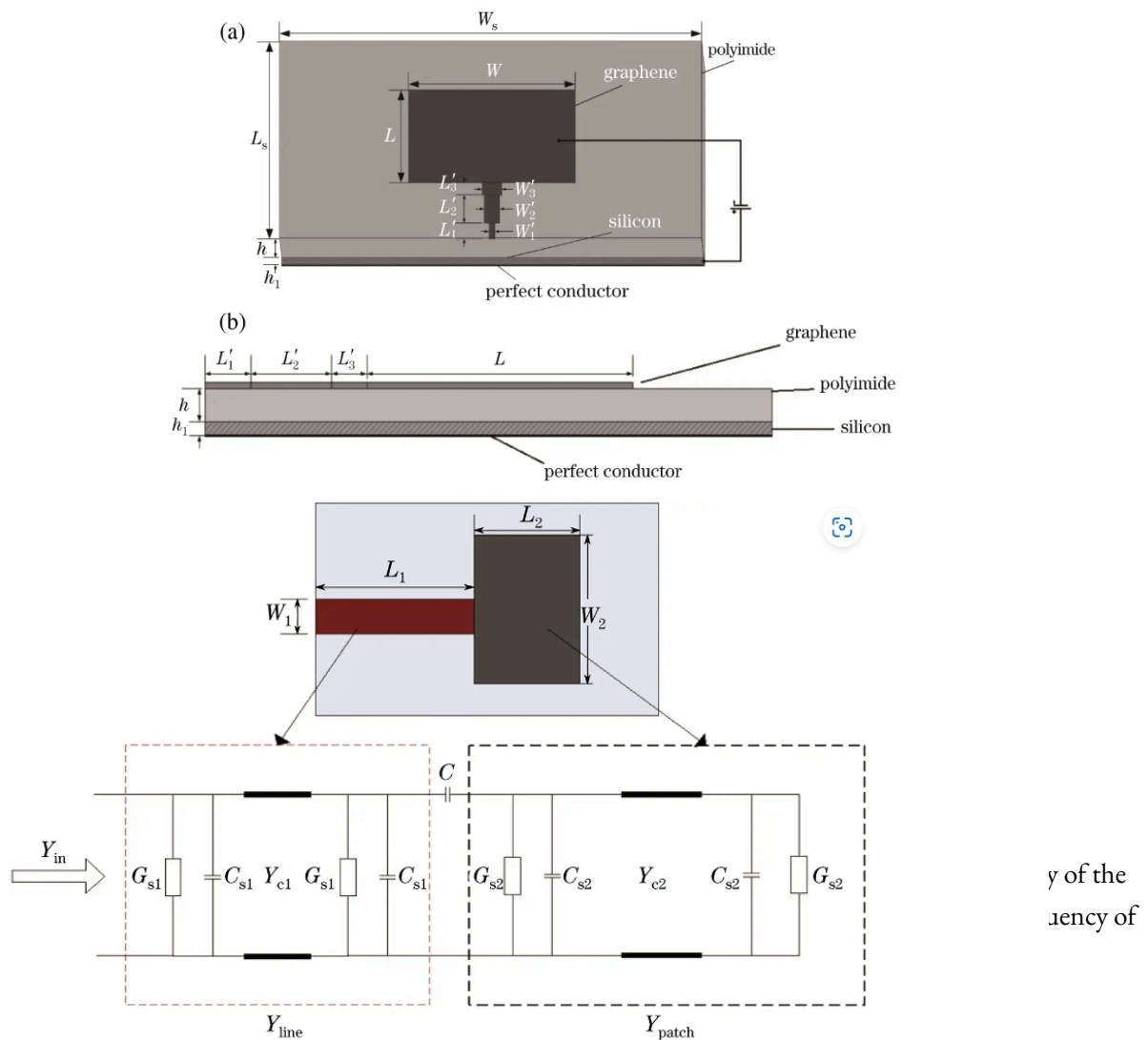
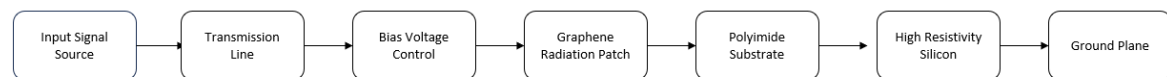


Figure 3: Equivalent circuit model of the proposed antenna

Bias Control Circuit: Controls the chemical potential of the graphene by adjusting the external bias voltage. By modifying the electric field across the graphene, this circuit dynamically tunes the resonant frequency.

Microstrip Transmission Line: This component matches the impedance between the THz source and the graphene patch. Proper impedance matching is critical in minimizing power loss and maximizing radiation efficiency.

Polyimide Substrate and Ground Plane: A polyimide substrate supports the microstrip and graphene patch, while the ground plane minimizes back radiation. Polyimide, with its low dielectric constant, reduces losses and enhances radiation efficiency.

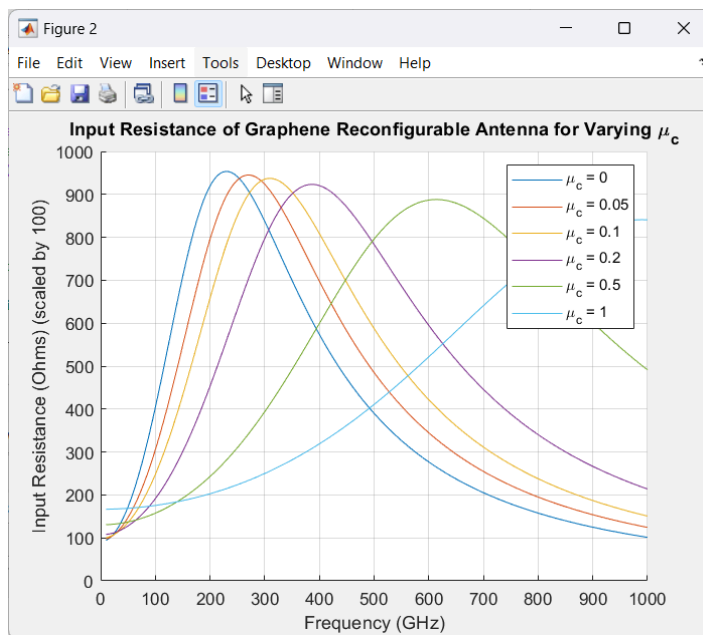
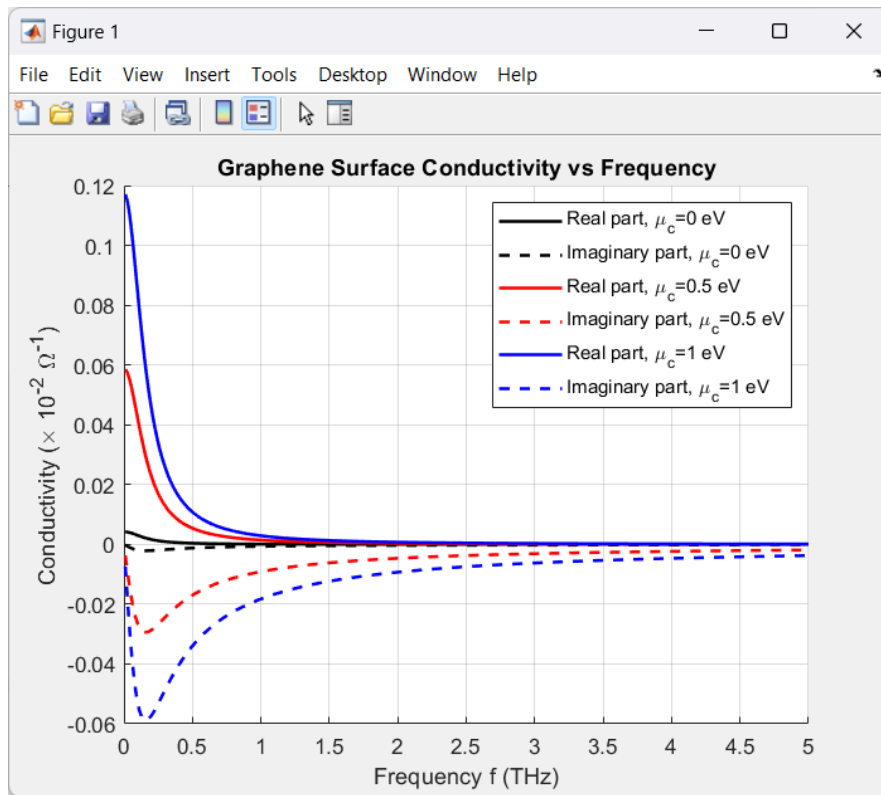
Implementation and Challenges: Each component was carefully simulated, with the bias control circuit being the most challenging aspect. It required accurate modeling of the graphene conductivity response to electric bias. The equivalent circuit model initially oversimplified coupling effects in the microstrip line, but further numerical simulations helped resolve this, ensuring better impedance matching.

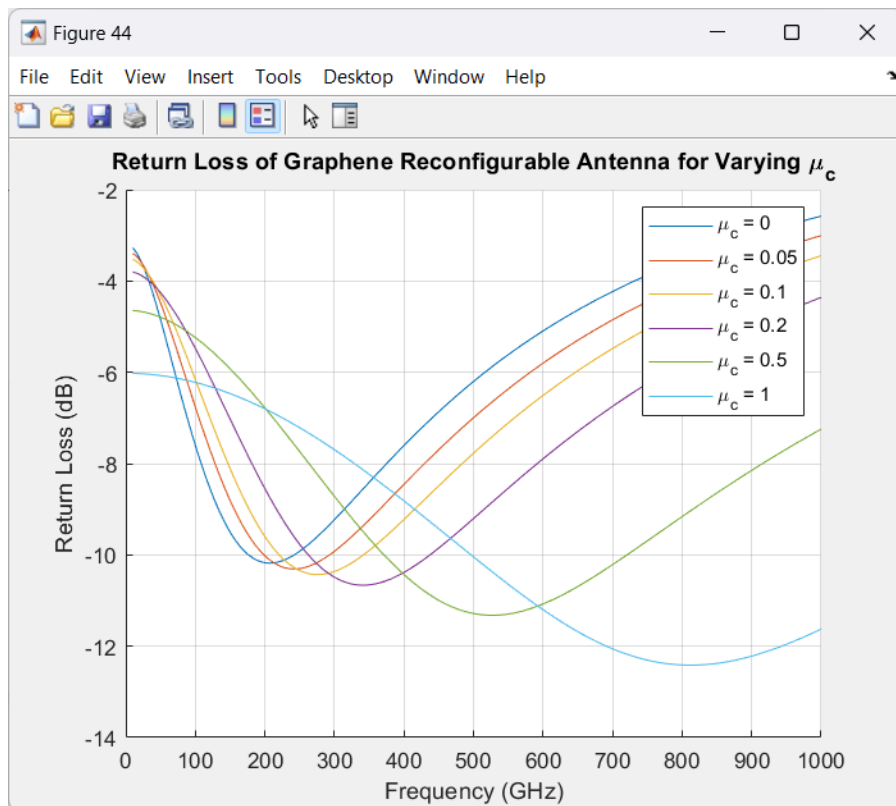
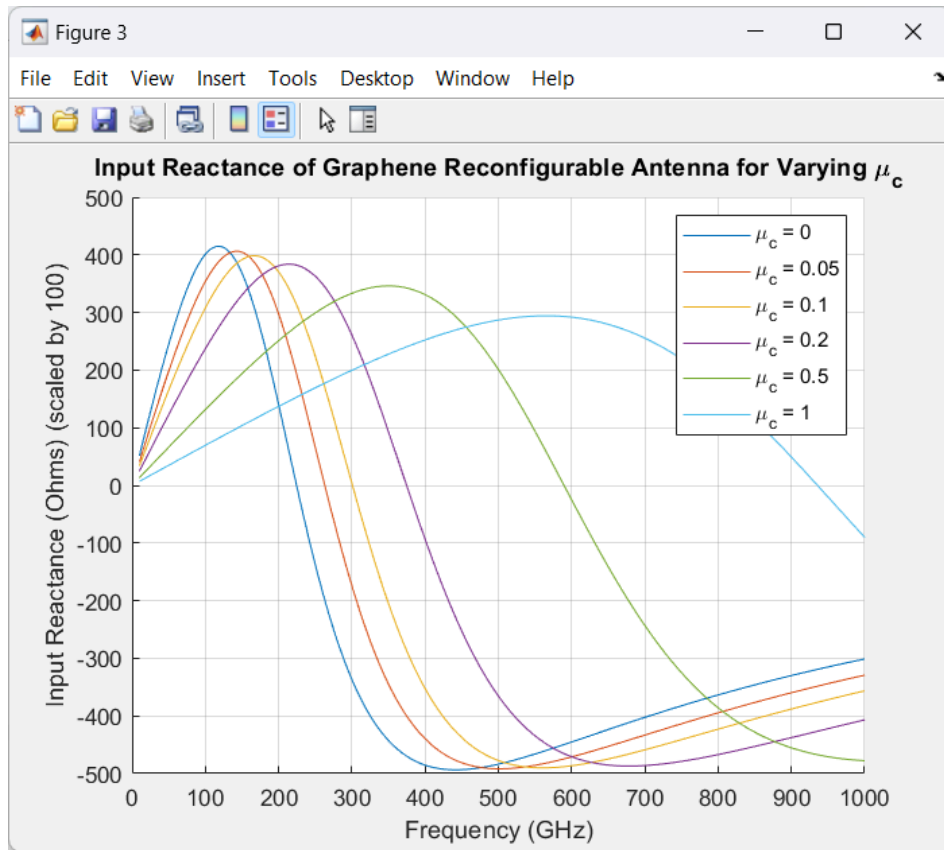
Demonstration of Results :

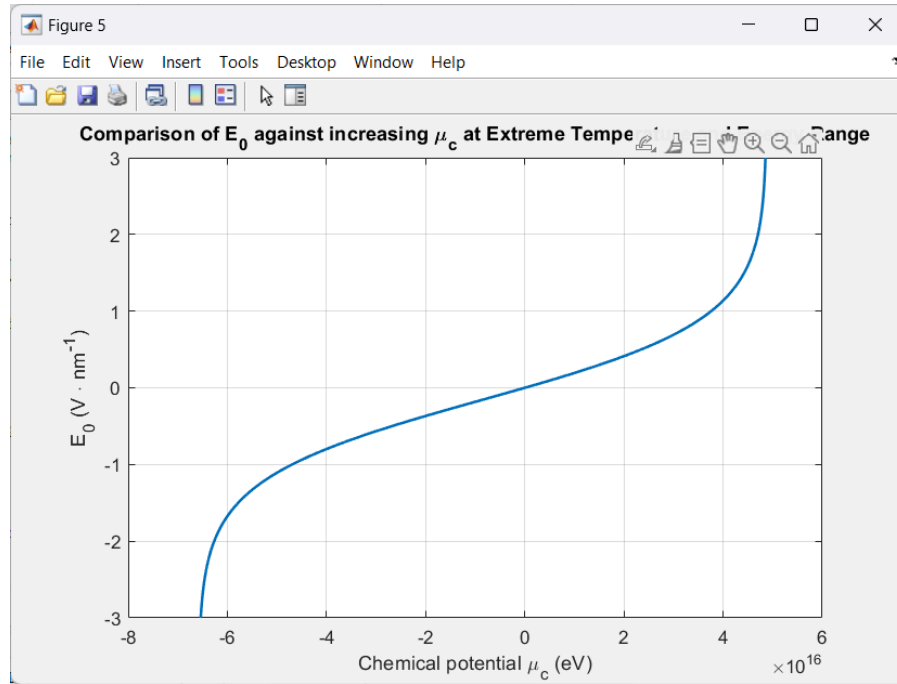
Flowchart for Results Demonstration:

1. Set Initial Parameters: Define initial graphene properties, chemical potential values, frequency range, and substrate characteristics.
2. Calculate Surface Conductivity: Using the Kubo formula, compute the real and imaginary parts of the graphene surface conductivity for various values of chemical potential.
3. Simulate Equivalent Circuit: Input impedance (Z_{in}) is computed for each frequency, factoring in the real and reactive components as defined by the equivalent circuit model.
4. Run Frequency Sweep Simulations: Simulate the antenna's performance at different frequencies across the range (695.55–705.05 GHz) to observe shifts in resonant frequency with changes in chemical potential.
5. Generate Plots and Analyze: Graph the real and imaginary parts of the impedance, surface conductivity, and return loss across frequencies for visual verification of the resonance and tunability.

Plots:







Supporting Plots and Analysis:

- **Input Impedance (Real and Imaginary):** Figures 2 and 3 in the document show that with increasing chemical potential, the input impedance decreases in the real part, while the imaginary part stabilizes at high frequencies, indicating better resonance matching.
- **Return Loss vs. Frequency:** Figure 7 demonstrates a return loss greater than 10 dB across multiple resonant points, verifying that the antenna achieves acceptable tuning across the specified THz range.
- **Radiation Pattern and Gain:** As shown in Figure 8, the radiation gain increases with chemical potential adjustments, achieving a maximum gain of 6.92 dBi. The radiation pattern aligns with the design requirements, maintaining high efficiency and directionality at each resonant frequency.

Conclusion :

The proposed graphene-based frequency reconfigurable antenna successfully achieves tunable resonant frequencies in the THz range (695.55 to 705.05 GHz) by leveraging graphene's dynamic conductivity. The antenna's tunability was validated through simulations, showing a smooth frequency shift with increasing chemical potential and bias voltage adjustments. Achieving a maximum radiation gain of 6.92 dBi and a radiation efficiency surpassing 86%, the antenna's performance met expectations, indicating its suitability for high-frequency THz applications in communication.

Future Work:

1. **Enhanced Frequency Range:** Future iterations could explore alternative substrates or hybrid materials with graphene to broaden the frequency range.
2. **Power Efficiency:** Further optimizing the bias control circuit could improve power efficiency, particularly by minimizing the tuning mechanism's energy consumption.
3. **Real-World Testing:** Experimental validation in real-world environments, especially under varying temperature conditions, would be essential for understanding practical applications in THz communications.

References:

Design and Analysis of a Novel Antenna for THz Wireless Communication, Omar A. Saraereh1,* , Luae Al-Tarawneh2, Ashraf Ali1 and Amani M. Al Hadidi, Intelligent Automation & Soft Computing
DOI:10.32604/iasc.2022.020216

R. Kaul, "Microwave engineering," *IEEE Potentials*, vol. 8, no. 2, pp. 11–13, 1989. [[Google Scholar](#)]

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Appendix:

Code:

```
% Constants
e = 1.602e-19;    % Elementary charge (C)
hbar = 1.055e-34; % Reduced Planck constant (J.s)
k_B = 1.381e-23;  % Boltzmann constant (J/K)
T = 300;          % Temperature (K)
tau = 1e-12;      % Relaxation time (s)
Gamma = 1/(2*tau); % Scattering rate (s^-1)
frequencies = linspace(0.01, 5, 500); % Frequency range (THz)
frequencies = frequencies * 1e12;    % Convert THz to Hz
% Chemical potentials
mu_c_values = [0, 0.5, 1] * e; % Chemical potential in eV converted to J
% Fermi-Dirac distribution function
f_d = @(epsilon, mu_c) 1 ./ (exp((epsilon - mu_c) / (k_B * T)) + 1);
% Intra-band conductivity formula
sigma_intra = @(omega, mu_c) - (1i * e^2 * k_B * T) ./ (pi * hbar^2 * (omega - 1i * 2 * Gamma)) .* ...
    (mu_c / (k_B * T) + 2 * log(1 + exp(-mu_c / (k_B * T))));
% Allocate arrays for results
real_sigma = zeros(length(frequencies), length(mu_c_values));
imag_sigma = zeros(length(frequencies), length(mu_c_values));
% Loop over chemical potentials and frequencies
for j = 1:length(mu_c_values)
    mu_c = mu_c_values(j);
    for i = 1:length(frequencies)
        omega = 2 * pi * frequencies(i);
        sigma = sigma_intra(omega, mu_c);
        real_sigma(i, j) = real(sigma); % Real part of conductivity
        imag_sigma(i, j) = imag(sigma); % Imaginary part of conductivity
    end
end
% Plotting
figure;
hold on;
colors = {'k', 'r', 'b'}; % Colors for different mu_c values
linestyles = {'-', '--'};
% Plot real and imaginary parts for each mu_c
for j = 1:length(mu_c_values)
    plot(frequencies / 1e12, real_sigma(:, j), 'Color', colors{j}, 'LineStyle', linestyles{1}, 'LineWidth', 1.5);
    plot(frequencies / 1e12, imag_sigma(:, j), 'Color', colors{j}, 'LineStyle', linestyles{2}, 'LineWidth', 1.5);
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end
% Graph details
xlabel('Frequency f (THz)');
ylabel('Conductivity ( $\times 10^{-2} \Omega^{-1}$ )');
legend({'Real part,  $\mu_c=0$  eV', 'Imaginary part,  $\mu_c=0$  eV', ...
    'Real part,  $\mu_c=0.5$  eV', 'Imaginary part,  $\mu_c=0.5$  eV', ...
    'Real part,  $\mu_c=1$  eV', 'Imaginary part,  $\mu_c=1$  eV'}, ...
    'Location', 'northeast');
grid on;
title('Graphene Surface Conductivity vs Frequency');
hold off;

% Clear previous data
clear;
clc;

% Constants
mu0 = 4 * pi * 1e-7;    % Permeability of free space (H/m)
sigma_v = 1e7;          % Bulk conductivity of graphene (S/m)
dg = 0.1e-6;            % Equivalent thickness of graphene (m)

% Reference impedance
Z0 = 50;                % Reference impedance (Ohms)

% Frequency range
f = linspace(1e8, 10e9, 1000); % Frequency from 100 MHz to 10 GHz
omega = 2 * pi * f;      % Angular frequency (rad/s)

% Antenna parameters
R0 = 500;                % Base Resistance (Ohms)
C0 = 5e-10;              % Base Capacitance (F)
L0 = 1e-3;               % Base Inductance (H)

% Define chemical potential values for mu_c
mu_c_values = [0, 0.05, 0.1, 0.2, 0.5, 1]; % Updated values for mu_c
num_values = length(mu_c_values);

% Initialize input impedance arrays and return loss array
Zin_real_all = zeros(num_values, length(f)); % Input resistance for each mu_c
Zin_imag_all = zeros(num_values, length(f)); % Input reactance for each mu_c
return_loss_all = zeros(num_values, length(f)); % Return loss for each mu_c

% Loop through each mu_c value
for m = 1:num_values
    mu_c = mu_c_values(m); % Current chemical potential

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% Calculate parameters based on mu_c
R_shift = 1 + mu_c;          % Resistance shift
C_shift = 5e-12 * (1 / (1 + 2 * mu_c)); % Capacitance shift
L_shift = 1e-9 / (1 + 5 * mu_c); % Inductance shift
% Initialize input admittance arrays for this mu_c
Yin_real = zeros(size(f));    % Real part of input admittance
Yin_imag = zeros(size(f));    % Imaginary part of input admittance
% Loop through frequencies to calculate input admittance
for i = 1:length(f)
    % Calculate total admittance
    Ysn = 0.1 + 1i * omega(i) * C_shift; % Equivalent admittance of the open slot
    Yline = 1 / (R_shift + 1i * omega(i) * L_shift); % Transmission line admittance
    Ytotal = Ysn + Yline;          % Total input admittance
    % Store real and imaginary parts
    Yin_real(i) = real(Ytotal);    % Real part of input admittance
    Yin_imag(i) = imag(Ytotal);    % Imaginary part of input admittance
end
% Calculate Input Impedance for current mu_c
Zin = 1 ./ (Yin_real + 1i * Yin_imag); % Input impedance
% Extract real and imaginary parts of input impedance
Zin_real_all(m, :) = real(Zin);    % Input resistance
Zin_imag_all(m, :) = imag(Zin);    % Input reactance
% Calculate Reflection Coefficient and Return Loss for current mu_c
reflection_coefficient = (Zin - Z0) ./ (Zin + Z0);
return_loss_all(m, :) = 20 * log10(abs(reflection_coefficient));
end
% Scale frequency and input impedance
f_scaled = f * 100;          % Scale frequency by 100
Zin_real_all = Zin_real_all * 100; % Scale input resistance by 100
Zin_imag_all = Zin_imag_all * 100; % Scale input reactance by 100
% Plot Input Resistance for different mu_c values in a new figure
figure;
hold on; % Hold on to plot multiple lines
for m = 1:num_values
    plot(f_scaled * 1e-9, Zin_real_all(m, :), 'DisplayName', ['\mu_c = ' num2str(mu_c_values(m))]);
end
grid on;
title('Input Resistance of Graphene Reconfigurable Antenna for Varying \mu_c');
xlabel('Frequency (GHz)'); % Label x-axis
ylabel('Input Resistance (Ohms) (scaled by 100)'); % Update label
xlim([0 1000]);          % Limit x-axis for better view
legend show;              % Show legend

```

```

hold off;
% Plot Input Reactance for different mu_c values in a new figure
figure;
hold on; % Hold on to plot multiple lines
for m = 1:num_values
    plot(f_scaled * 1e-9, Zin_imag_all(m, :), 'DisplayName', ['\mu_c = ' num2str(mu_c_values(m))]);
end
grid on;
title('Input Reactance of Graphene Reconfigurable Antenna for Varying \mu_c');
xlabel('Frequency (GHz)'); % Label x-axis
ylabel('Input Reactance (Ohms) (scaled by 100)'); % Update label
xlim([0 1000]); % Limit x-axis for better view
legend show; % Show legend
hold off;
% Plot Return Loss for different mu_c values in a new figure
figure;
hold on; % Hold on to plot multiple lines
for m = 1:num_values
    plot(f_scaled * 1e-9, return_loss_all(m, :), 'DisplayName', ['\mu_c = ' num2str(mu_c_values(m))]);
end
grid on;
title('Return Loss of Graphene Reconfigurable Antenna for Varying \mu_c');
xlabel('Frequency (GHz)'); % Label x-axis
ylabel('Return Loss (dB)'); % Label y-axis
xlim([0 1000]); % Limit x-axis for better view
legend show; % Show legend
hold off;
% Display message to confirm that separate figures were created
disp('Separate figures generated for Input Resistance, Input Reactance, and Return Loss.');
```

```

% Constants
e = 1.60217662e-19; % Elementary charge in Coulombs
h_bar = 1.054571817e-34; % Reduced Planck constant in J.s
v_F = 1e6; % Fermi velocity in m/s
epsilon_b = 4; % Example dielectric constant of the medium
k_B = 1.380649e-23; % Boltzmann constant in J/K
% Set extreme temperature
T = 12000; % Extreme temperature in Kelvin
% Fermi-Dirac distribution function
```

```

f_d = @(epsilon, mu_c) 1 ./ (exp((epsilon - mu_c) / (k_B * T)) + 1);
% Increase upper limit for energy integration to an extreme value
energy_limit = 1e-19; % Very large upper integration limit
% Integrand function for E0
integrand = @(epsilon, mu_c) (f_d(epsilon, 0) - f_d(epsilon + 2 * mu_c, 0));
% Function to calculate E0 with numerical integration
E0 = @(mu_c) (e / (pi * h_bar^2 * v_F^2 * epsilon_b)) * integral(@(epsilon) integrand(epsilon, mu_c), 0,
energy_limit);
% Range of chemical potentials (in eV)
mu_c_eV = linspace(-3, 3, 500); % Chemical potential in eV
mu_c_J = mu_c_eV * e; % Convert to Joules
% Calculate E0 for each chemical potential
E0_values = arrayfun(E0, mu_c_J);
% Plotting the results
figure;
plot(mu_c_eV, 1);
plot(E0_values, mu_c_eV, 'LineWidth', 1.5);
xlabel('Chemical potential \mu_c (eV)');
ylabel('E_0 (V \cdot nm^{-1})');
title('Comparison of E_0 against increasing \mu_c at Extreme Temperature and Energy Range');
grid on;

```